

# MSc Computing Project Research Proposal and Ethical Approval Form

## Student Details

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|-----------------------------------|--------------------------------------------------------------------------------------------|
| <b>Student name</b>               | Todd Edge                                                                                  |
| <b>Proposed title of research</b> | To explore how an LLM-powered interactive pedagogical agent can support autistic educators |
| <b>Supervisor</b>                 | Anupam Mazumdar                                                                            |

## Section 1 Research Proposal

**Provide a Research Proposal which addressed the following (in no more than 2000 words):**

1. An introduction to the problem and the motivation behind your project, including your research question.
2. The project aims and its supporting objectives.
3. A skeleton of your literature review, highlighting key sections and the underlying topics.
4. An overview of the research methods that you will use for data collection and analysis.
5. A breakdown of your project into work packages, including a description of the artefact(s) to be created and your timeline.
6. For Cyber Security projects only, please indicate the CyBOK category and knowledge area applicable to your project.

Remember to use UoEO Harvard referencing. A Gantt chart or similar can be used for your timeline.

### **Background**

Autistic educators face many workplace challenges that have led to many leaving the profession due to their subsequent poor mental health (StEvens, 2024; Jang et al., 2024; Dumonteil, 2022; Wood & Happé, 2023; O'Neill & Kenny, 2023). Unsupportive colleagues, ambiguous and secretive practices, and unsuitable working environments have been cited amongst the reasons for AUTISTIC educators struggling in the workplace (StEvens, 2024; Jang et al., 2024; Dumonteil, 2022; Wood & Happé, 2023; O'Neill & Kenny, 2023). Research has shown that such educators' strengths often get overlooked despite the fact that there is strong evidence to suggest that they can be an invaluable resource for supporting students, particularly those with autism, acting as key role models (StEvens, 2024; Jang et al., 2024; Dumonteil, 2022; Wood & Happé, 2023; O'Neill & Kenny, 2023). Their unique way of thinking has also been found to be of vital importance to creativity within education, while their alternative perspective on education has been said to offer great potential when utilised within staff development sessions (StEvens, 2024; Jang et al., 2024; Dumonteil, 2022; Wood & Happé, 2023; O'Neill & Kenny, 2023).

Educational institutions, and the often neurodivergent people who oversee them, often disable autistic educators through neglecting to see autism as a condition that leads to a different way of thinking, which should be accommodated by society (Wood & Happé, 2023; Papadopoulos, 2024); instead, autism is often seen as a problem to fix, which has caused many to mask their symptoms, leading to adverse effects to their mental health (StEvens, 2024; Jang et al., 2024; Dumonteil, 2022; Wood & Happé, 2023; O'Neill & Kenny, 2023). Indeed, many educators do not disclose their autism diagnosis for fear of judgement and unfair treatment (StEvens, 2024; Jang et al., 2024; Dumonteil, 2022; Wood & Happé, 2023; O'Neill & Kenny, 2023). While LLMs can offer vital support to autistic adults, challenges in using them include the technology utilising biased data, hallucinations, financial barriers, and LLMs seeing autism in a simplified way, perpetuating stereotypes (Papadopoulos, 2024).

### **Introduction**

Research on the utilisation of large language models (LLMs) for supporting autistic individuals in education has overwhelmingly focussed on students. Although studies have been undertaken to uncover the benefits and potential of utilising LLMs to support autistic adults (Lara, Lacave & Molina, 2025; Haroon & Dogar, 2024; Jang et al., 2024; Hrabal, Davis & Wicker, 2022), there is a distinct lack of research on how LLM agents can be used to support autistic educators.

Haroon & Dogard (2024) developed an application to help its autistic users to understand social complexities inherent within text messages. This application had features such as a meaning interpreter, emotional response previewer and response suggester to help its users to navigate these challenges. The researchers found that their users wished for a more collaborative experience, where suggestions were made on how to improve responses, rather than wholesale changes being made (Haroon & Dogard, 2024). The authors also acknowledged the need for the application to explicitly state – within the user interface - how the application was to be used as a guide, aiming to prevent users' over-relying on it as a social tool (Haroon & Dogard, 2024).

Indeed, Jang et al. (2024) found that LLMs are often sought out by autistic adults to assist them with challenges such as social interactions, however these have been found to output unhelpful responses, due to the models being trained on neurotypical data. Although autistic users' were responses positively praised LLMs, autistic expert practitioners warned that many of its responses were ungrounded and overly optimistic (Jang et al., 2024). Jang et al.'s (2024) study raises questions around the complexities of offering advice to a vulnerable group – which disability model should such an LLM reflect: a Social Model that sides with the Standpoint Theory that social structures should be challenged by those with disabilities, or The Medical Model that advises that participants should conform to social and workplace norms. Papadopoulos (2024) similarly suggests that current LLMs are biased towards a one-dimensional medical model of disability where autism is something to be corrected.

Papadopoulos (2024) and Pryor et al. (2025) suggest the need to educate developers in disability-informed ethics, while encouraging autistic individuals to be invited to co-design applications. Pryor et al, (2025) further suggest grounding LLM-based applications in reliable data on autism. Retrieval augmented generation (RAG) has been used to ground LLM-based applications in reliable datasets that have reduced the likelihood of erroneous, hallucinatory outputs whilst offering more flexibility and contextually sensitive responses (Alsafari et al., 2024; Lewis et al., 2020).

These research findings will be utilised in the development of an LLM application to assist autistic educators in their daily practice.



Within the proposed study, with the help of 7 autistic educators, I will seek to co-design, develop, test and evaluate a research-informed pedagogical LLM agent for autistic educators – SpectrumAI. This LLM agent will seek to offer emotionally aware professional support to enable its users to maintain good mental health and uphold positive educational practice. The LLM will have a customisable setting that can tailor responses based on a different disability model –The Social Model, based around the concept of empowering the individual and challenging the social structure around them, and the Medical Model, built on the idea that disabled individuals should be supported to manage external challenges and conform to social settings (Jang et al., 2024). Depending on the customisation configuration, it can offer support in 3 main ways:

- It will offer supportive suggestions to help autistic educators to decide on next steps after experiencing workplace stressors such as a lack of support, and social politics.
- It will assist with lesson planning and resource generation by suggesting ways to draw upon and celebrate the user's monotropism – their intense focus and passion for specific subjects or topics.
- It will help the educators to find opportunities to model to autistic pupils how they navigate their environment with autism.

The research question for this study is:

- To what extent can a customisable LLM-powered interactive pedagogical agent – SpectrumAI – support autistic educators?

### **Project aims and objectives**

The project aim is to **develop an LLM-powered interactive pedagogical agent – Spectrum AI – that offers effective emotionally intelligent, supportive assistance to autistic educators**. The following objectives will support this aim:

- To identify the issues and challenges facing autistic educators in schools.
- To design and develop, with 7 autistic educators, an interactive LLM-powered pedagogical agent - SpectrumAI.
- To test and implement Spectrum AI with the 5 educators.



## **Literature review skeleton**

### **Introduction**

- LLMs in education
- LLMs in Special Educational Needs and Disabilities (SEND) education
- Autistic Educators
- LLMs for autistic adults

■

### **Theme A: LLMs in SEND education**

- Communication
- Personalisation
  - Dynamic LLM communication based on neurodivergent profile (Nalli, Kapetanakis & Nguyen, 2025).
- Accessibility
  - Adaptable, flexible design, preserving self-authenticity (Jamshed et al., 2025).
- Emotional

### **Theme B: Experiences of autistic educators**

- Challenges
  - Stereotypes
  - Masking/lack of disclosure of diagnosis
  - Double empathy problem
  - Social politics in schools
  - Environment
  - Lack of support
  - Executive functioning
- Strengths
  - Strong connection with autistic students and marginalized students
  - Creative problem solving
  - Resource for leading teachers in understanding how to cater for autistic students
  - Monotropism

### **Theme C: LLMs for autistic adults**

- Planning
  - Visual - shared calendar, collaborative plan management (Lara, Lacave & Molina, 2025).

- Communication
- Limitations of LLMs for assisting autistic adults (Papadopoulos, 2024)
  - Biased training data – trained on neurotypical information
  - Over-reliance
  - Misinformation, stigmatizing language, hallucinations, especially in free models
  - Financial barrier of paid models
  - Data security
  -
- Disability models
  - The Medical Model: tech to help manage or mitigate challenges
  - The Social Model & Standpoint Theory: tech should empower user and challenge society (Jang et al., 2024).
  - Neurodiversity model (Papadopoulos, 2024)
  -
- Future directions
  - The Path Forward – ethical LLM development for disabled individuals (Papadopoulos, 2024)
  - RAG (Pyror et al., 2025; Alsafari et al., 2024; Lewis et al., 2020)
  - Awareness & education – LLMs as assistants rather than an tool for deception.
  - Collaborative development: developer engagement with autistic individuals; address bias in training data.
  - Enforce privacy and data protection
  - Positive representation

### **Research method overview**

SpectrumAI will be developed in collaboration with 7 autistic educators, informed by design principles recommended for LLMs developed for neurodivergent individuals (Jang et al., 2024; Papadopoulos, 2024; Jamshed et al., 2025; Nalli, Kapetanakis & An Nguy, 2025; Haroon & Dogar, 2024). The development process will follow an Agile Scrum methodology, with feedback provided from the teachers' review at the end of each sprint. Acceptance testing will be conducted with the 5 educators. An evaluation of the agent will be conducted through a questionnaire and semi-structured interview with the 7 autistic educators.

The methodology of collaborative development is based on suggestions made in research on ethical LLM application development (Papadopoulos, 2024; Pryor et al., 2025). Using an agile methodology will best allow for collaboration, fostering redesign and a customer-focused product.

The following problems may arise:

Participants drop out

RAG does not solve the problem of erroneous/unhelpful advice

It will be more difficult to get participants to test the application online – they will need guidance in the use of the technology

RAG does not solve the problem of erroneous/unhelpful advice

**Limitations:**

Ideally, a separate group of people would review the application.

The number of autistic educators available for participation is low.

**Artefact to be created**

An LLM agent tailored towards supporting autistic educators - SpectrumAI.

**Features:**

**RAG architecture**, utilising a knowledge base of reliable, research-informed data on autism.

**Transparent choice:** user has an explicit choice between output based on these the Medical Model or Social Model of disability (Jang et al., 2024), outlining the risks of both e.g. actions taken as a result of the advice based on the Social Model could result in negative responses from others.



### **Work packages (WP)**

#### **WP1: Literature review**

Search for appropriate literature, make notes then write-up.

#### **WP2: Design questionnaire and interview**

Design the questionnaire and interview to gather data post-development on educators' evaluation of the system.

#### **WP3: Secure volunteers' consent**

Ensure 7 autistic teachers are willing to participate and obtain informed consent through written contract.

#### **WP4: SDLC (Scrum)**

Requirements gathering/analysis with autistic teachers; System design: using storyboards, class diagrams, flow charts; Develop the system: code the back end (Python) and front end (Tkinter); Testing: test the code with unit and manual tests; Deployment: deploy the application; Review: autistic teachers to review and feedback; Repeat all of the above until teachers are satisfied that requirements have been met.

#### **WP5: Questionnaire evaluation**

Obtain data through questionnaire to 7 teachers

#### **WP6: Conduct interviews**

Conduct group interview to gain further information on opinions on the system.



## WP7: Data analysis

Statistical analysis, interpret results

## WP8: Write-up

Write up methodology (including images of application), results, discussion, conclusion

## WP9: Submit dissertation

Submit final copy of dissertation

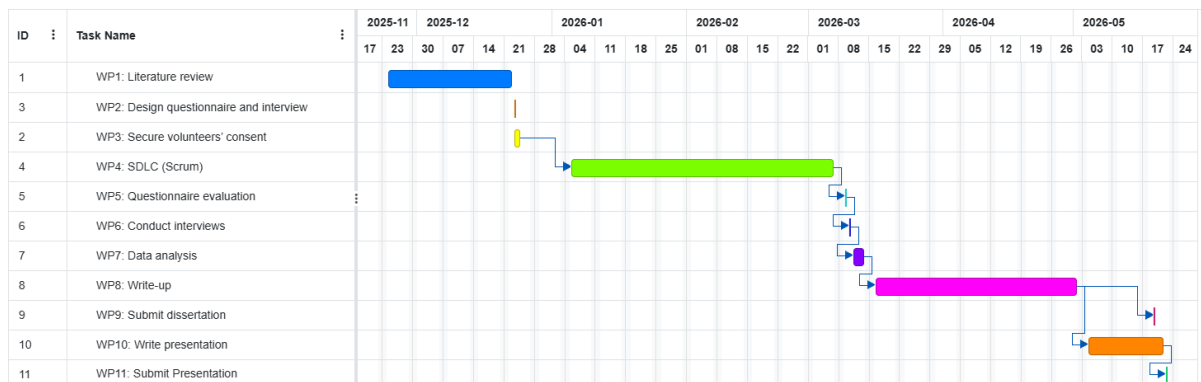
## WP10: Write presentation

Draft, redraft and finalise dissertation presentation

## WP11: Submit Presentation

Submit final copy of presentation

## Timeline



## **References**

Alsafari, B., Atwell, E., Walker, A. & Callaghan, M. (2024) 'Towards effective teaching assistants: From intent-based chatbots to LLM-powered teaching assistants', *Natural Language Processing Journal*, 8, 100101. Available at: <https://doi.org/10.1016/j.nlp.2024.100101> (Accessed 21 November 2025)

Dumonteil, J. (2022) 'Autistic teachers: the culmination of an inclusive journey?', *Revue Education et Socialisation*, (65).

Haroon, R., & Dogar, F. (2024, October). Twips: A large language model powered texting application to simplify conversational nuances for autistic users. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility* (pp. 1-18).

Hrabal, J. M., Davis, T. N. & Wicker, M. R. (2023) 'The use of technology to teach daily living skills for adults with autism: A systematic review', *Advances in Neurodevelopmental Disorders*, 7(3), pp. 443-458. Available at: <https://doi.org/10.1007/s41252-022-00255-9> (Accessed: 20 November 2025).

Jamshed, H., Naseem, M., Potluri, V. & Brewer, R. N. (2025) 'Rethinking Productivity with GenAI: A Neurodivergent Students' Perspective', *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*. pp. 1-16. Available at: <https://doi.org/10.1145/3663547.3746329> (Accessed: 13 November 2025).

Jang, J., Moharana, S., Carrington, P. & Begel, A. (2024) "'It's the only thing I can trust": Envisioning large language model use by autistic workers for communication assistance', *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. pp. 1-18. Available at: <https://doi.org/10.1145/3613904.3642894> (Accessed: 29 November 2025).

Lara, J., Lacave, C. & Molina, A. I. (2025) 'PlanTEA-WM: A Multi-User Web Platform for Routine Planning and Anticipating Everyday Situations in Individuals with Autism Spectrum Disorder', *IEEE Access*, 13, pp. 180523-180538. Available at: 10.1109/ACCESS.2025.3617152 (Accessed: 28 November 2025).

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N. & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33, 9459-9474.

Nalli, G., Kapetanakis, S. & Nguyen, K. A. (2025) 'AI intelligent tutoring system tailored to the students' personality and neurodiversity', in *Proceedings of the 24th European Conference on e-Learning*, Vol. 24, No. 1, pp. 297-305.

O'Neill, C. & Kenny, N. (2023) "'I saw things through a different lens...': An Interpretative Phenomenological Study of the Experiences of Autistic Teachers in the Irish Education System', *Education Sciences*, 13(7), 670. Available at: <https://doi.org/10.3390/educsci13070670> (Accessed: 21 November 2025).

Papadopoulos, C. (2024) 'Large language models for autistic and neurodivergent individuals: Concerns, benefits and the path forward', *Neurodiversity*, 2. Available at: 10.1177/27546330241301938 (Accessed: 27 November 2025).

Pryor, K., Coleman, T., Hossain, S. Z., Tootil, E., & Ahmed, S. (2025, July). Examining Large Language Models Within Autism-Related Contexts: A Systematic Review of Bias and (Mis) Representation. In *2025 IEEE 49th Annual Computers, Software, and Applications Conference (COMPSAC)* (pp. 876-886). IEEE.

S Stevens, C. (2024) 'The lived experience of autistic teachers: a review of the literature', *International Journal of Inclusive Education*, 28(9), pp. 1871-1885. Available at: <https://doi.org/10.1080/13603116.2022.2041738> (Accessed: 23 November 2025).

Wood, R. & Happé, F. (2023) 'What are the views and experiences of autistic teachers? Findings from an online survey in the UK', *Disability & Society*, 38(1), pp. 47-72. Available at: 10.1080/09687599.2021.1916888 (Accessed: 24 November 2025).

## Section 2 Ethical approval

Use the following headings to describe and discuss how you intend to deal with ethical issues in relation to your research. You should consult the university guidelines and project module content to help you to do this. Acknowledge how you propose to address the issues if your research or part of your research is going to take place online. (**No more than 2000 words**)

### 1. Consent

How do you intend to seek informed consent from participants?

With a signed written contract.

I will let participants know about their role within the study (to co-design, test and review).

A copy of the consent form is attached.

### 2. Right to withdraw

How do you intend to inform participants of their right to withdraw?

Within the consent form. I will also mention this at the start of each online meeting, sharing my email and phone number should they wish to contact me to withdraw at any time.

### 3. Confidentiality

How do you intend to maintain confidentiality?

I will not use real names at any point during research documentation; I will refer to the educators as Educator A, Educator B etc.

### 4. Harm

How do you intend to protect participants from harm?

I will meet with participants online rather than in-person.

Potential risks to subjects include being misled by the LLM application in terms of advice offered.

To minimise this risk, I will regularly inform participants that the LLM application is a work-in-progress and could output unhelpful advice errors or even erroneous data. This warning will also be included in the participant consent form.

## 5. Data access, storage and security

Please confirm that all personal data will be stored and processed in compliance with the General Data Protection Regulation (GDPR). Describe the arrangements for storing and maintaining the security of any personal data collected as part of the project.

The collection of personal data will be kept to a minimum. Only initials will be recorded, along with phone numbers and email addresses. This data will be stored on on my personal laptop, as a password-protected 7Zip file and encrypted using AES-256.

Online design discussions will be recorded. Recordings will be stored on on my personal laptop, as a password-protected 7Zip file and encrypted using AES-256.

All personal data relating to the participants will be deleted after the submission of the dissertation.

## 6. Other issues

Identify any specific ethical issues relating to this research, for example if your research involves vulnerable groups like young children, or pupils who have SEND (special educational needs/disability).

When testing and reviewing the developed system, the educators will be told only to use initials of children or adults referred to in a chat. As the system will be powered by a ChatGPT model, using its API, the chats will be encrypted in transit (using TLS 1.2+) and at rest (using AES-256 encryption on OpenAI's servers). Any chats completed are not used to train the generic model.

As some of the participants have not disclosed their autism diagnosis to their employer, the participants' diagnoses will be kept confidential and not revealed to employers.



## Section 3 Risk Assessment

**If your research does not involve human participants, you are able to enter "N/A" in the comment box.**

1. Are there any potential risks, for example physical, psychological, social, legal or economic, to participants or subjects associated with the proposed research?

YES

Please provide full details of the potential risks and explain what risk management procedures will be put in place to minimise the risks:

Meetings are online so no on-premises risk assessment to abide by

**Hazard:** The application's output

**Hazardous event:** Social/psychological: SpectrumAI may output inaccurate/biased/masking advice.

**Consequence:** Impact on participants' mental health if advice followed.

**Risk management procedures:** Participants will be informed of the this within the written consent form. The potential for inaccurate/biased/masking advice will be detailed on the participant consent form as well as the application's graphical user interface, where participants will be reminded not to follow its advice while it is in development. In future iterations of the application, the LLM will be grounded (using retrieval augmented generation) in reliable data on autism in order to further reduce the risk of such output.

2. Are there any potential risks to researchers as a consequence of undertaking this proposal?

NO

Please provide details and explain what risk management procedures will be put in place to minimise this.

3. Are there any potential reputational risks to the University of Essex Online as a consequence of undertaking this proposal?

NO

Please provide full details and explain what risk management procedures will be put in place to minimise this.

4. Will the research involve individuals below the age of 18 or individuals of 18 years and over with a limited capacity to give informed consent?

NO



(If yes, a Disclosure and Barring Service disclosure (DBS check) may be required. Please attach as part of your application). Give further details of participants below.

5. Are there any other ethical issues that have not been addressed, which you would wish to bring to our attention?

NO

Give details below:

## Section 4 Confirmation Statements

The results of research should benefit society directly or by generally improving knowledge and understanding. I confirm that my research project has a potential benefit. (If you cannot identify a benefit, you must discuss your project with your supervisors to help identify one or adapt your proposal so the study will have an identifiable benefit.) /

I confirm that I have read the Research Ethics Policy and the relevant sections of the Research Ethics Procedures and will adhere to these in the conduct of this project. /

(These statements must be ticked in the form.)

## Signature



**Todd Edge**

**15.12.25**

## Attachments

You are required to attach the following documents to this form:

1. An example of your participant information sheet and consent form, if applicable.
2. Consent document from the organisation your research is taking place, if applicable. N/A